Paleomagnetic Zones in the Oligocene East Sooke Gabbro, Vancouver Island, British Columbia

D. T. A. SYMONS[1]

Geological Survey of Canada, Ottawa, Ontario, Canada

The 39-m.y.-old East Sooke polyphase gabbro stock intrudes Late Eocene Metchosin volcanics, shows no evidence of metamorphism, and is unconformably overlain by undeformed Late Oligocene sediments. A grid of samples (36 sites, 180 cores, 360 specimens) was collected in and around its 5- × 2½-mile (8 × 4 km) outcrop area at the southernmost end of Vancouver Island. Antiparallel directions, a positive contact test, thermomagnetic curves, and so forth indicate that all phases after af demagnetization retain a characteristic magnetization residing in magnetite. The stock contains a central zone of normal polarity bounded by two reverse zones. The zones are independent of petrologic boundaries. Thus all phases were solidified above the determined Curie temperatures (505° ± 10°C). Gradational direction trends in the zones show the irregular migration of isotherms in the stock while the earth's magnetic field switched polarity at least once and probably twice. The East Sooke pole position (150.9°E, 69.5°N, $\delta_m = 6.7°$, $\delta_p = 5.3°$) reflects a 20° ± 7° counterclockwise twist rotation of the southern Tertiary belt relative to the rest of Vancouver Island.

The East Sooke and Rocky Point gabbro stocks are located at the southernmost end of Vancouver Island (Figure 1). During June 1969, cores were collected at nine sites (sites 28–36 in Figure 2) from these stocks as part of a paleomagnetic study of Tertiary igneous rocks on the island [*Symons*, 1971*b*]. Curiously the East Sooke gabbro was found to contain normal and reverse zones of characteristic remanent magnetization after partial alternating field (af) demagnetization within its 10-mi² (25 km²) area. Thus additional sites (sites 1–27) were collected in 1970 to define more clearly the zones of the stock and to examine their significance.

GEOLOGY

In their discussion of the regional geology of the Sooke and Duncan map areas *Clapp and Cooke* [1917] give a more detailed description and geologic map (1:24,000 scale) of the East Sooke and Rocky Point stocks. These stocks are two of several gabbroic intrusions known as the Sooke intrusives found in a belt of Tertiary rocks on the south end of Vancouver Island. This belt (Figure 1) is about 35 miles (55 km) long and reaches a maximum width of 10 miles (16 km) between the Leech River thrust fault to the north and the Strait of Juan de Fuca to the south. Besides unconsolidated Pleistocene and Recent deposits, there are three Tertiary rock units in the belt: the Late Eocene Metchosin volcanics, which are intruded by the Early Oligocene Sooke intrusives, and the Late Oligocene Sooke clastic sediments, which unconformably overlie both older units. Along the Leech River thrust fault the Tertiary units have been overthrust by metamorphosed late Paleozoic units known as the Malahat chlorite-quartz-feldspar schists of volcanic origin and the Leech River slaty schists of sedimentary origin and by Jurassic island intrusions composed of granitic gneisses [*Muller*, 1971].

The Metchosin volcanic sequence has an estimated thickness of 7500 feet (2400 meters). It is composed of fine-grained tholeiitic basalts, some of which exhibit pillow or columnar structures. It includes some porphyritic and amygdaloidal flows and interbedded agglomeratic units that yield Late Eocene fossils. The sequence has an average strike of N70°W and a dip of 20°NE, some variation being due to folding. The fold axes trend parallel to the strike, i.e., N70°W; thus the folding was caused by compression associated with the overthrusting of the Mesozoic units. The sequence is broken by

[1] Now at Department of Geology, University of Windsor, Windsor, Ontario, Canada.

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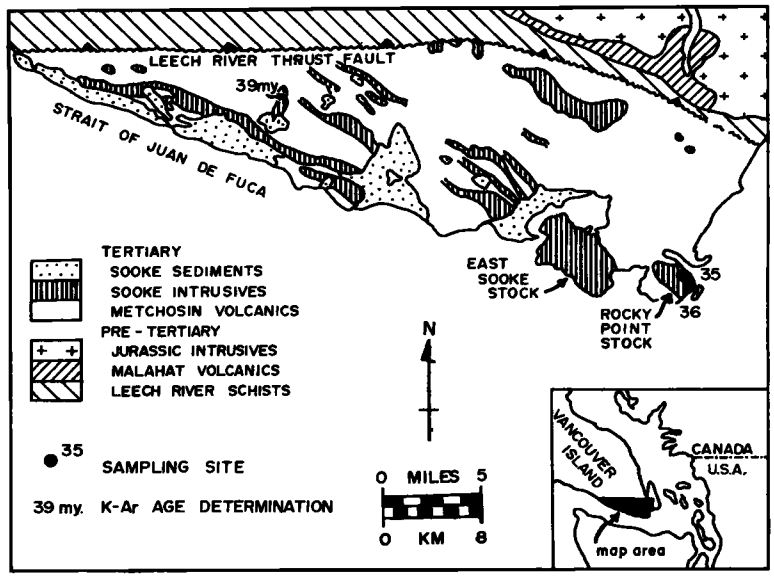


Fig. 1. Location and general geology of the southern Vancouver Island Tertiary belt after *Clapp and Cooke* [1917] and *Carson* [1969], and location of the Rocky Point stock sampling sites.

numerous nearly vertical shear and fault zones that strike rudely parallel to the Leech River thrust fault. Cross shears are most commonly found around the Sooke intrusive masses.

The Sooke intrusives occur as numerous elongate stocks trending parallel to the fold axes of the Metchosin volcanics [*Carson*, 1969]. They are composed of gabbro with minor anorthositic and granitic phases. One intrusive near the west end of the Tertiary belt yields an Early Oligocene K-Ar age of 39 m.y. [*Wanless et al.*, 1967; *Holmes*, 1960], which would be Upper Eocene in some more recent Cenozoic time scales [*Berggren*, 1969]. Adjacent to the stocks the Metchosin volcanics have been contact metamorphosed, strongly fractured, and cut and replaced by veinlets of quartz, epidote, and other minerals.

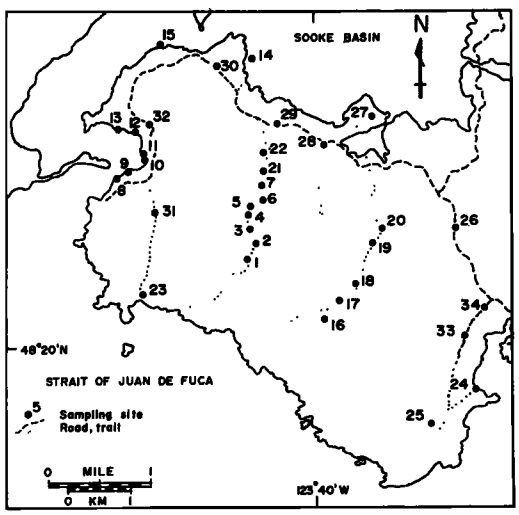


Fig. 2. Location of East Sooke gabbro stock sampling sites.

The East Sooke stock is the largest and best exposed of the Sooke intrusives. It is 5 miles (8 km) long and 2½ miles (4 km) wide, thereby forming most of the East Sooke Peninsula (Figure 3). The stock has a core of olivine gabbro surrounded in large part by a border phase of augite gabbro next to its contact with the Metchosin volcanics. Anorthositic and granitic phases are very locally present. From other studies on ultramafic bodies this sort of zoning is apparently common [*Taylor*, 1967] and indicates that the augite gabbro crystallized first with the olivine gabbro, the anorthosite and granite phases following as later differentiates. Phase contacts are generally gradational although sharp contacts have been observed. A number of gabbroic dikes, both porphyritic and nonporphyritic, are present. They are cut by a few later granitic dikes and veins composed of

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hornblendite and aplite. The veins are usually associated with vertical faults with small horizontal displacements. More specifically the hornblendite veins carry minor sulfide mineralization, which has been described by *Fyles* [1949] and *Carson* [1969]. Very locally the gabbroic phases show differentiation banding and flow structures. The gabbro is unaltered in hand sample and thin section except in the ⅛-inch-thick weathering rind, along joints, and in the shear zone. Magnetite is the only magnetic mineral reported in the gabbro.

The Sooke sediments unconformably overlie and occur in structural basins between ridges of the Metchosin basalts and Sooke intrusives. The sediments are composed of a basal conglomerate overlain by a sequence of sandstone and conglomerate with a few interbeds of shale, marl, and lignite. This sequence of nonresistant predominantly marine sediments rarely exceeds a few hundred feet in thickness. It strikes generally parallel to the coast, dips at less than 5°S, and was evidently deposited in a shallow marine and coastal plain environment. The Sooke sediments contain Late Oligocene and very Early Miocene fossils [*Sutherland Brown*, 1966].

SAMPLING

In this study, five core samples, each oriented in situ by sun compass, were obtained from each of 36 collecting sites in or near the East Sooke gabbro stock (Figure 2). The sites are distributed as uniformly as practicable over the stock. Of these sites, 31 are from within the outcrop limits of the East Sooke stock (Figure 3), of which 23 sites (sites 1–8, 11–13, 16–22, 25, and 31–34) are from the olivine gabbro phase, 6 sites (sites 15, 23, 26, and 28–30) are from the augite gabbro phase, 1 site (site 9) is

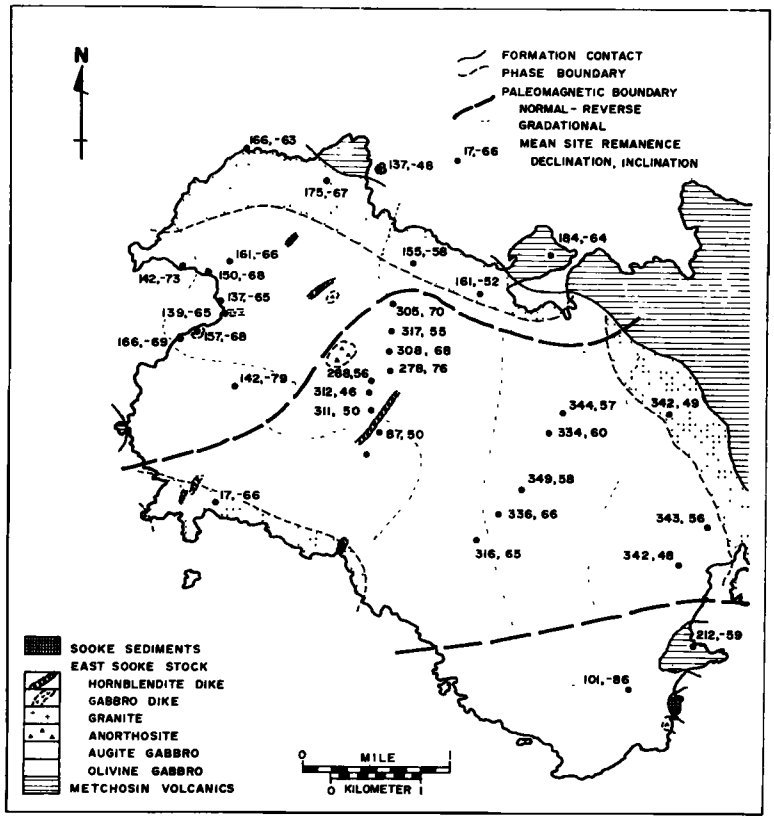


Fig. 3. Geology of East Sooke gabbro stock after *Clapp and Cooke* [1917] and *Fyles* [1949], site mean remanence directions, and resultant paleomagnetic zones.

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TABLE 1. Geologic, af Step Demagnetization, and Remanent Magnetization Data by Sites

Site	Geology[a]	af Step Demagnetization			Reject[e]		Site Mean Remanence[f]						
		SI[b]	Range[c]	Field[d]	i.m.	a.r.m.d.	n'	R	D	I	k	α_{95}	SE
1	OG	31.9	300 to 500	400	2		3	2.5495	119.6	74.3	4	38.5	22.2
2	OG	7.4	300 to 500	400			5	4.8492	87.1	49.8	27	12.2	7.0
3	OG	16.2	500 to 800	650			5	4.9742	311.4	50.3	155	5.0	2.9
4	OG	9.0	500 to 800	650	3		2	1.9943	312.1	46.2	174	7.5	4.3
5	OG	17.2	200 to 500	400			5	4.9717	287.9	55.7	141	5.3	3.0
6	OG	16.4	100 to 400	200	1		4	3.9194	278.1	76.1	37	11.5	6.7
7	OG	25.1	0 to 200	100	2		3	2.9833	307.7	68.3	119	7.4	4.3
8	OG	27.8	100 to 400	200			5	4.9897	166.4	-69.0	390	3.2	1.8
9	A	30.1	200 to 400	300			5	4.9982	156.8	-68.2	2260	1.3	0.8
10	PGD	14.1	200 to 650	500	2		3	2.9899	139.0	-64.8	199	5.7	3.3
11	OG	30.1	400 to 800	500			5	4.9839	136.9	-64.6	249	4.0	2.3
12	OG	26.9	100 to 800	500			5	4.9913	150.1	-67.9	457	2.9	1.7
13	OG	11.4	0 to 300	300			5	4.9880	142.2	-73.3	333	3.4	2.0
14	MV	1.8	100 to 400	200	1	1	3	2.9593	137.2	-48.2	49	11.6	6.7
15	AG	25.5	200 to 500	300			5	4.9900	165.7	-63.1	401	3.1	1.8
16	OG	20.4	500 to 800	650			5	4.9262	316.3	65.4	54	8.5	5.0
17	OG	32.2	50 to 500	100	1		4	3.9866	335.6	65.7	223	4.7	2.7
18	OG	16.7	200 to 800	400	1		4	3.9922	349.1	57.6	386	3.6	2.1
19	OG	19.2	100 to 800	200	1	1	3	2.9991	334.1	59.5	2199	1.7	1.0
20	OG	3.7	200 to 400	300	2		3	2.9967	344.0	56.6	600	3.3	1.9
21	OG	27.8	0 to 400	400			5	4.9968	317.4	55.3	1246	1.8	1.0
22	OG	28.3	0 to 200	100	3		2	1.9970	305.0	70.0	337	5.4	2.6
23	AG	4.0	0 to 400	200		1	4	3.9518	17.0	65.8	62	8.9	5.2
24	MV	1.1	400 to 650	500	1	1	3	2.9358	211.8	-58.7	31	14.5	8.4
25	OG	54.1	100 to 300	200	1		4	3.9821	100.5	-85.6	168	5.4	3.2
26	AG	25.3	100 to 500	300			5	4.9611	342.2	48.7	103	6.2	3.6
27	MV	2.1	200 to 400	300	1	1	3	2.9520	183.6	-64.3	42	12.6	7.3
28	AG	15.7	100 to 400	200	1	1	3	2.9956	160.9	-52.1	459	3.8	2.2
29	AG	21.5	0 to 400	200			5	4.9723	155.1	-57.7	144	5.2	3.0
30	AG	40.0	50 to 800	200			5	4.9960	174.7	-66.6	992	2.0	1.2
31	OG	5.4	300 to 800	300	1	1	3	2.9714	141.7	-78.9	70	9.7	5.6
32	OG	20.1	200 to 800	300			5	4.9981	161.3	-65.6	2113	1.4	0.8
33	OG	17.9	200 to 400	300		1	4	3.9838	342.1	47.7	185	5.2	3.0
34	OG	20.6	50 to 800	100			5	4.9948	343.3	56.0	765	2.3	1.3
35	AGR	37.9	200 to 400	300		1	4	3.9626	204.5	-21.7	80	7.8	4.5
36	AGR	35.9	0 to 800	100			5	3.8712	189.0	-47.3	23	14.5	8.4

[a]Unit sampled: MV, Metchosin volcanics. Sooke intrusions: OG, olivine gabbro; AG, augite gabbro (AGR, Rocky Point stock); A, anorthosite; and PGD, porphyritic gabbro dike.
[b]Stability index value.
[c]Range of minimum and maximum demagnetizing field in oersteds.
[d]Intensity of demagnetizing field used for site in oersteds.
[e]Number of cores rejected as inhomogeneously magnetized (i.m.) or having anomalous remanent magnetization direction (a.r.m.d.).
[f]Parameters: n', number of cores used to calculate site mean remanence data; R, length of resultant vector giving each core unit weight; D, I, declination and inclination of resultant vector in degrees; k, Fisher's dispersion statistic; α_{95}, radius of cone of 95% confidence; and SE, radius of cone of standard error in degrees.

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from the anorthosite phase, and 1 site (site 10) is from a porphyritic gabbro dike. Two sites (sites 35 and 36) from augite gabbro represent the Rocky Point stock, which is located about 3 miles (5 km) ESE of the East Sooke stock (Figure 1). Finally, 3 sites (14, 24, and 27) are from contact-metamorphosed Metchosin volcanics around the perimeter of the East Sooke stock (Figures 2 and 3).

PROCEDURE

Two specimens were cut from each core, giving a total of 360. After measuring the natural remanent magnetization (NRM) of each, one pilot specimen was selected to represent each site on the basis of its having an average NRM direction and intensity for the site. The pilot specimen was step demagnetized in an alternating field with peak intensities of 50, 100, 200, 300, 400, 500, 650, and 800 oe, and its remanent magnetization was remeasured after each step. By using the stability index (SI) method [*Tarling and Symons*, 1967] and remanence intensity decay curves as a guide, an optimum af intensity was selected for each site. The remaining specimens from each site were then demagnetized (Table 1) in this optimum field before remeasuring their remanence. The biastatic magnetometer and af demagnetizing apparatus used have been described by *Larochelle and Christie* [1967] and *Larochelle and Black* [1965], respectively.

In 24 cores the cleaned remanence directions of the two specimens diverged by more than 20°; such divergence reflects an inhomogeneous magnetization, and so the results from these cores were rejected. The mean remanent magnetization direction of each of the remaining cores was then calculated from the average of its two specimens. In nine sites, one core direction was found to deviate from the mean direction of the three or four remaining cores by more than 3 times their circular standard deviation, and so the results from these nine cores were likewise rejected. The site mean remanence directions were then calculated (Table 1) by giving each remaining core direction unit weight. Finally one site (site 1) was found to have a large cone of 95% confidence ($\alpha_{95} \geq 15°$), which reflects inhomogeneous remanent magnetization at the site, and so its site mean remanence is not considered further.

DISCUSSION

Remanence stability. There is excellent evidence that the characteristic remanent magnetization of the gabbroic rocks of the East Sooke stock is highly stable and primary in origin. (1) The rocks show no thermal or dynamic alteration during which a secondary remanence may have been introduced. (2) The af demagnetization tests give an average SI value of 21.0, which indicates an extremely stable remanence. (3) The radius of the average cone of 95% confidence (α_{95}) about the site mean directions is low (5.3°), indicating stable site magnetization. (4) High field susceptibility versus temperature measurements on samples from five sites (sites 4, 6, 8, 34, and 35) including both normal and reversely magnetized sites and both olivine and augite gabbro phases indicates that magnetite with a Curie temperature of 505° ± 10°C is the sole ferromagnetic mineral (e.g., Figure 4*a*).

Conversely the evidence suggests that the Metchosin volcanics adjacent to the East Sooke stock contain unstable remanence of secondary origin. These rocks show clear petrographic evidence of both contact thermal and dynamic metamorphism, indicating that secondary remanence components are likely. Their SI values give an average of 1.7, indicating unstable remanence. Their average α_{95} is 12.9° after

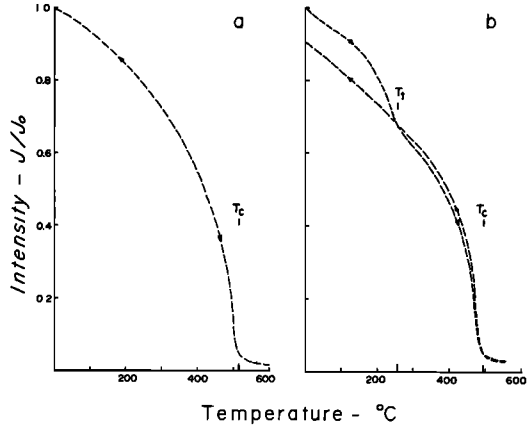


Fig. 4. Thermomagnetic curves obtained during heating and cooling of samples from (*a*) East Sooke olivine gabbro, site 34 (J_o = 1.45 emu/g) and (*b*) metamorphosed Metchosin volcanics, site 14 (J_o = 2.22 emu/g); T_c is the Curie temperature; T_t is the maghemite–hematite transition temperature. The applied field is 5200 oe.

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rejection of two cores from each of the three sites, indicating inhomogeneous site magnetization. Closer examination reveals (Figure 5) that each site had (1) one core rejected as stable but anomalous, which had a remanence direction moderately inclined to the north, (2) one core rejected as inhomogeneously magnetized, which had inconsistent reversed inclinations, and (3) three cores accepted, which gave a cluster of reversed remanence directions to the south that are reasonably close to the direction obtained for the nearest site within the East Sooke gabbro. Finally the high field susceptibility versus temperature curve for a sample from site 14 (Figure 4*b*) indicates the presence of both magnetite and maghemite.

This evidence is interpreted as follows. The Metchosin volcanics had a stable primary remanence of normal polarity directed to the north. When the East Sooke gabbro was intruded, the volcanics were thermally altered, the result being a change in their ferromagnetic mineralogy. During intrusion they acquired a somewhat more stable and intense secondary remanent magnetization component parallel to that of the adjacent gabbro. Therefore the volcanics now contain a dominant secondary component and a recessive primary component. These results provide a contact test that supports the other cited evidence indicating the presence of a primary stable remanence in the East Sooke gabbro stock.

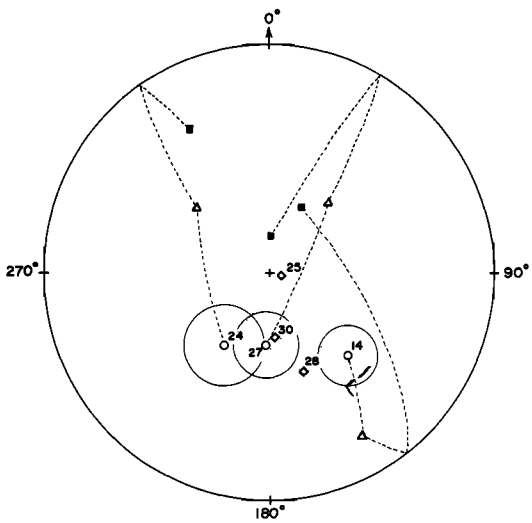


Fig. 5. Remanence directions from the Metchosin volcanics. The site mean direction (circles) with its cone of 95% confidence reflects the dominant secondary component, and it is linked (dotted lines) to the nearest East Sooke stock site (diamonds). The mean is also linked (dashed lines) to the core rejected as inhomogeneous (triangles) with mixed components and then to the core rejected as anomalous (squares), which reflects the recessive primary component. Solid and open symbols indicate normal and reversed inclinations, respectively.

Remanence zoning. The normally and reversely magnetized sites from the East Sooke stock occur in zones unrelated to the petrologic phases (Figure 3). All sites from the northwestern end of the stock have similar reversed directions despite the fact that they represent four phases, i.e., olivine gabbro, augite gabbro, anorthosite, and porphyritic gabbro dike. Through the middle of the stock there is a normally magnetized zone representing both the olivine and the augite gabbro phases. Conversely the southeastern end of the stock contains reversed remanence directions from the olivine gabbro phase and, if the satellite Rocky Point stock some 3 miles to the southeast is included, from the augite gabbro phase as well. Therefore all phases must have been emplaced and solidified before any of the intrusion cooled below the Curie temperatures (≈505°C) of the magnetic minerals in any of the rocks.

Within each polarity zone there appear to be distinct trends in the remanence directions that cannot be explained tectonically. For example, in the northwestern reverse zone going from SW to NE, there are 2 sites with inclinations equal to or greater than −70°, then 8 sites with inclinations between −60° and −69°, and finally 2 sites with inclinations equal to or less than −59°. Similarly, in the middle normal zone, 2 sites in the SW have declinations of 17° and 87°. In the NW and central portion there are 8 sites with declinations between 278° and 317°. In the eastern 7 sites the declinations are between 334° and 349°, inclinations being between 57° and 66° on the west side and between 48° and 56° on the east side. Further, site 25 in the southeastern reverse zone and the two sites (sites 35 and 36) from Rocky Point give directions distinctly different from those in the northwestern reverse zone. These zones and trends suggest that the isotherms during cooling through the rocks' Curie points migrated through the stock in an irregular fashion while

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the polarity of the earth's magnetic field switched from reverse to normal and back to reverse. Conveniently there was a series of very short term (i.e., about 30,000 year) polarity changes about 39 m.y. ago (Figure 6). Alternatively the zones could reflect some short undetected polarity event.

One evident conclusion to be drawn from these polarity and remanence trends is that even relatively small and stably magnetized intrusive stocks must be thoroughly sampled to obtain a reliable mean remanence direction and derived pole position.

Stock mean remanence direction. The site mean remanent magnetization directions of the 32 valid sites from the Sooke intrusives form two distinct clusters on the stereonet (Figure 7). One cluster of normal polarity is inclined at 60° ± 10° to the NNW, and the second has an antiparallel inclination to the SSE. This antiparallelism is additional evidence in favor of the gabbro retaining a stable primary remanence. However, there are three sites that do not conform to either cluster. Within the East Sooke stock, site 2 deviates noticeably in direction and is the sole site collected adjacent to a mineralized shear zone. Although no alteration is evident in the cores from site 2, its remanence may have been altered during the mineralizing episode. The other two sites (sites 35 and 36) are from the Rocky Point stock. The deviation of their magnetization directions from the norm may have been caused by factors such as a difference in time of intrusion from the East Sooke stock, an aberrant ambient earth's magnetic field during a polarity transition, or some undiscovered intervening tectonic event that has caused rotation.

Data from sites 2, 35, and 36 were therefore excluded, and the remaining 29 sites from within the East Sooke stock were then analyzed by giving each site mean remanence direction unit weight. Two tests were made to assure that the directions form a homogeneous population. First, the reversed remanence directions were rotated through 180° to the normal position, and the angular variance test was applied to see if the normal and reversed populations can be combined [*Larochelle*, 1968]. Because the calculated statistic ($\delta_b^2/\delta_w^2 = 2.69$) is less than the theoretical statistic for the 95% confidence level ($F_{2,54,0.95} = 3.17$), the two populations may

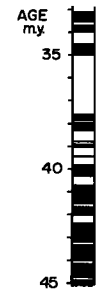


Fig. 6. Reversal chronology from 33 to 45 m.y. age after *Berggren* [1969] with black and white areas indicating times of normal and reverse magnetic polarity, respectively.

be validly combined. The same result is obtained when the olivine gabbro sites are compared to the sites from all other phases ($\delta_b^2/\delta_w^2 = 3.01$; $F_{2,54,0.95} = 3.17$). When the 29 sites are combined, the resulting stock mean remanence direction is 329.9°, 64.2° (Fisher statistics: $R = 28.2809$; $k = 38.9$; $\alpha_{95} = 4.2°$). By means of the stock mean remanence direction and the dipole formula, the resulting pole position for the East Sooke stock is 150.9°E, 69.5°N ($\delta_m = 6.7°$; $\delta_p = 5.3°$).

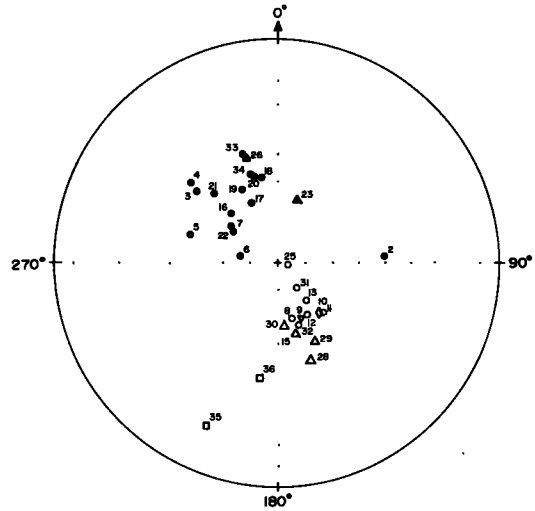


Fig. 7. Site mean remanence directions for the Sooke intrusives. East Sooke stock: olivine gabbro (circles), augite gabbro (triangles), anorthosite (reversed triangle), and porphyritic gabbro dike (diamond). Rocky Point stock: augite gabbro (squares). Solid and open symbols indicate normal downwardly and reversed upwardly directed inclinations, respectively.

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TABLE 2. Paleomagnetic Pole Positions from the North American Plate from Oligocene-Eocene

Pole	Age	Unit	Pole Position		δ_m, deg	δ_m, deg	Reference
			Long., °W	Lat., °N			
1	Oligocene	west central U.S. extrusives	214	78	11	9	*Grommé and McKee* [1971]
2	Oligocene	Mt. Washington dacites	157	64	10	10	*Symons* [1971*b*]
3	Oligocene	East Sooke gabbro stock	209	70	7	5	This paper
4	Eocene	Wasatch red beds	180	84	27	20	*Torreson et al.* [1949]
5	Eocene	Green River red beds	158	78	7	6	*Torreson et al.* [1949]
6	Eocene	Green River red beds	168	85	9	8	*Torreson et al.* [1949]
7	Eocene	Hope granitic pluton	152	88	6	5	*Symons* [1973]
8	Eocene	Siletz River volcanics	49	37	10	7	*Cox* [1957]

The pole numbers are used in Figure 8; δ_m, δ_p, semiaxes of the oval of 95% confidence.

Significance of the pole position. Only a few paleomagnetic pole positions have been reported for Eocene-Oligocene units from the North American craton (Table 2). *Grommé and McKee* [1971] have reported a pole position for a sequence of lavas and tuffs from the west central United States. *Torreson et al.* [1949] have reported positions for the Wasatch and Green River red beds that appear to reflect a primary magnetization despite their not being magnetically cleaned. All these poles are essentially concordant and are located midway in years between the present pole, i.e., <1.0 m.y. [*Wilson*, 1971], and the mean Cretaceous pole for the North American craton [*Beck and Noson*, 1972] as befits their intermediate age (Figure 8).

It is apparent that the poles for Eocene-Oligocene units in the western cordillera are not necessarily concordant with the cratonic poles. The Hope granitic complex from the northern Cascade Mountains [*Symons*, 1973] gives a concordant pole. The discordancy found for the Mount Washington dacites of Vancouver Island is not unexpected, because the sampled stock and flows are probably comagmatic and may be coeval. Thus secular variation may not have been truly averaged out, giving a virtual pole position [*Symons*, 1971*b*]. Finally the Siletz River volcanics [*Cox*, 1957] of Oregon give a pole position in the mid-Atlantic east of Bermuda. This discordant result has been attributed in a geologically consistent way to a 70° clockwise post-Eocene rotational deformation [*Carey*, 1958; *Wise*, 1963; *Irving*, 1964].

The East Sooke gabbro pole is consistent with the cratonic poles in remanence inclination to within about 7° but deviates in declination by 20° ± 7°. The discrepancy in the East Sooke gabbro pole position could be caused tectonically by a 12° ± 5°E tilt rotation about a horizontal N-S axis or by a 20° ± 7° counterclockwise twist rotation about an adjacent vertical axis. The overlying Late Oligocene Sooke sediments are flat lying, so that any tilt would have had to occur immediately after intrusion. The intruded Metchosin volcanics strike N70°W and dip 20°NE on the average. If the volcanics and stock were tilted after intrusion, the East Sooke pole would be 'corrected' to mid Russia (075°E, 65°N), which is totally incompatible with its Eocene age. Clearly, as the geologic relations indicate, the Metchosin tilt was primarily preintrusive. Also, there is no evidence

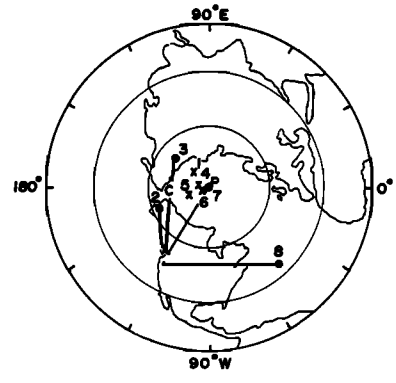


Fig. 8. Polar projection of the northern hemisphere showing Oligocene-Eocene paleomagnetic poles. Poles for rock units in the western cordillera are shown as solid circles joined by lines to their sample sites. Poles from units in the rest of the North American craton are shown as crosses. The pole numbers refer to Table 2. The present (<1.0 m.y.) pole of *Wilson* [1971] and mean Cretaceous pole for the North American craton [*Beck and Noson*, 1972] are indicated by P and C, respectively.

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for a 12° ± 5°E tilt rotation of the stock. There is some geologic evidence for a 20° ± 7° counterclockwise twist rotation of the southern end of Vancouver Island, including the Tertiary belt. J. E. Muller (personal communication, 1971] has pointed out that the formational trends and major faults at the island's southern end strike N65° ± 5°W, whereas throughout most of the length of the island to the north they trend N40°W [*Geological Survey of Canada,* 1968]. If the trends and faults were originally aligned, the southern end has been relatively rotated 25° ± 5° counterclockwise. Such rotation is compatible, though not unequivocally, with compressional movement along the Leech River thrust fault and the hypocenter cluster reported in the Puget Sound–Juan de Fuca Strait area seismic data [*Tobin and Sykes,* 1968]. Further, it is consistent with the rational explanation of the pole position found for the mid-Jurassic island intrusions [*Symons,* 1971*a*]. They were extensively sampled throughout the length of Vancouver Island north of the Leech River fault. Their pole position shows no statistically significant postintrusive movement of the island or insular belt relative to cratonic North America. If a statistically nonsignificant movement has occurred, there are two geologically probable possibilities. (1) A 10° ± 5° clockwise twist rotation is possible. Significantly, if the formational and fault trends on Vancouver Island were corrected to a N50°W alignment, both the East Sooke gabbro pole and the island intrusion poles would become essentially concordant for their respective ages. (2) A few degrees of NE tilt rotation associated with extensive block faulting are also possible, which may account for the slightly steeper than expected remanence inclination.

The discordancy between the East Sooke gabbro and cratonic North American poles can be explained in part (i.e., for remanence inclination) by a 10° northward translation of the gabbro relative to the craton. On the basis of very limited paleomagnetic data, *Beck and Noson* [1972] and *Irving and Yole* [1972] contend that the western cordillera has moved north relative to the rest of the North American plate by 25° of latitude since the Cretaceous and 50° since the Triassic. The translation hypothesis does require the 20° ± 7° counterclockwise twist rotation of gabbro relative to craton to correct for declination discrepancy. Conversely both the contemporaneous Mount Washington dacite pole position (when it is assumed that secular variation has been averaged) and the mid-Jurassic island intrusions pole position would require a 10° southward translation plus twist rotation of Vancouver Island to be rationalized by the translation mechanism.

CONCLUSIONS

The 39-m.y.-old East Sooke gabbro stock on the southernmost end of Vancouver Island retains a primary characteristic remanent magnetization. The stock contains a central remanence zone of normal polarity bounded on either side by reversed zones. These zones are thought to reflect rapid polarity changes in the paleofield; such change is consistent with the known reversal chronology. The zones are independent of the petrologic phase boundaries, so that their internal gradational changes in remanence direction are thought to reflect the irregular migration of isotherms through the stock during cooling. These systematic directional variations emphasize the importance of extensive sampling in paleomagnetic studies even when relatively small structures composed of magnetically stable rock types are studied.

The pole position obtained for the East Sooke stock (150.9°E, 69.5°N, $\delta_m = 6.7°$, $\delta_p = 5.3°$) is discordant with the few pole positions reported for coeval rocks from the North American craton. This discordance is caused by a 20° ± 7° counterclockwise twist rotation of the Tertiary belt at the southern end of Vancouver Island relative to the rest of the island to the north and to cratonic North America.

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